

# COSC 2436

## Sections:

### Structure of course

- What will you learn over this course?
- When will you learn certain topics?
- Is there a late work policy? - Kenny
  - There is no late work policy. Every due date is final. Because of the transparency of deadlines throughout the course for HWs, CIAs, CMAPS, Dlabs, and Glabs, it is expected that students submit work on time or receive a score of 0. Additionally, most due dates provide students with weeks at a time to complete the assignment.

### Homework 0 - Joseph

- ArgumentManager usage in homeworks
- Testing locally on Windows, Mac.
  - Via server
  - Via terminal (Mac)
  - Via terminal (Windows)
- Uploading homework onto server
- Checking grade of homework, and failed test cases, alongside what each file generated means – ie. .diff, etc...
- How do regrades work - Kenny
  - To submit a regrade, update your homework directory (folder) on the server, using FileZilla or any other FTP software, with the new program files before the time constraints listed below.
  - Note that after each initial homework deadline, a hw#-archive folder will be created and available on the [2436.cs.uh.edu](http://2436.cs.uh.edu) server, where you can view the hidden test cases for expected outputs. This should allow you to test your solution locally and ensure your program functions as intended before resubmitting for a regrade.
  - After the initial homework deadline has passed, you can check your grade via Putty and/or your file transfer protocol service (FileZilla or other alternatives). If you are not satisfied with your homework grade, you have up to 24 hours to submit a second attempt. After that 24 hour period has ended, you can then again check your grade - by viewing not the initial hw folder, but hw#-regrade1. If you are still unsatisfied, you will have another 24 hours to resubmit a final attempt. This time, check your grade within the hw#-regrade2 folder.
  - In summary, after each initial deadline for homework assignments, you have up to 48 hours to submit up to 2 regrade attempts, separated by 24 hours. You may choose to only attempt one regrade of your choice. However, keep in mind that

the reward for completing the regrade within 24 hours as opposed to 48 is greater. See the detailed score breakdown below.

- The first regrade is the original score + (the score after 1st regrade - original score) \* 80%. For example, if your original grade was a 40 and then you got an 80 after the first regrade, then your grade after the first regrade would be  $40 + (80 - 40) * 80\% = 72$ .
- The second regrade is the score after the first regrade + (score after the second regrade - score after the first regrade) \* 60%. For example, if your grade after the first regrade was a 72 (as in the example above) and then you got a 100 on the second regrade, your final homework grade would be  $72 + (100 - 72) * 60\% = 88$ .
  - If you only attempted the second regrade, then the “score after the first regrade” will simply be your initial grade for that homework.
- Your final homework grade can be calculated by doing:  $(\text{original\_grade} + (\text{1st\_regrade\_grade} - \text{original\_grade}) * 80\%) + (\text{2nd\_regrade\_grade} - \text{grade\_after\_1st\_regrade}) * 60\%$
- Final Regrade information - Kenny
  - Towards the end of the semester, after all homework assignments have been due, students will have the opportunity to replace one homework's grade entirely. In order to be eligible for the regrade, a student must resubmit an attempt for all homework assignments that were not already at a 100. The homework that sees the highest improvement will then be adjusted to reflect the newest attempt's score. Note that only one homework score will be replaced, regardless if two or more see the same level of improvement. Improvement is defined by the difference in the two grades.
- How do I change permissions of a file?
- What do I need to upload on the server?
- How do I login, and what is my login?
- Who do I contact if I need help with the server?

## CIA - Truman

- How to enter a group with other students?
  - In order to enter in groups with other students, you will need to head over to your courses and select COSC 2436. Then, select the people tab and select groups. Once you are in this tab, you then choose one of the following available groups. If you don't have anyone in mind for your groups, you can always join any open one! However, if you already have your group in mind, choose a group with 0/3 people inside. If you ever feel like switching to other groups, inform the TA's and follow their directions.
- How many questions/How to prompt
  - Each CIA must have a total of 21 questions. Of these 21 questions, there must be 10 Conceptual Questions, which include understanding theory, principles, and key concepts in programming, and 10 algorithm based questions, which are problem-solving and implementation-focused questions. There should be 5 MCQ,

5 Fill In The Blank, 5 T/F, and 5 Word Answers included in the 10 conceptual and algorithm based questions. After all of the following requirements have been met, you must create a 21st question which states "Rate this quiz done by Group X. Select one option only." This question should have 1-10 as the answer choices.

- How to generate them?
  - Use ChatGPT to generate your questions. This is to help you understand the subject more as Chat will give you factual information and good questions in order to understand the topics correctly. If you don't know how to prompt GPT for the questions, you could ask GPT to "create 20 questions total, 10 must be Conceptual and the other 10 must be algorithm based. Of these 10 conceptual and 10 algorithm based questions, Include 5 Fill In The Blank, 5 T/F, and 5 Word Answers" Be sure to include the topics.
- How many have to be done, and include placeholders for due dates.
  - There must be 2 CIA's completed before the semester ends. CIA 1 will be due on June, 16 at 11:59 pm. CIA 2 will be due on June, 23 at 11:59 pm.
- What is each CIA related to?
  - CIA 1 will include arrays, lists, introduction & analysis of algorithms, and doubly linked lists. CIA 2 will include stacks, bubble sort, selection sort, insertion sort, priority queues, and heaps.
- How do you upload them and test their syntax?
  - You will need to go to NYIT Canvas Exam Converter. Then read the directions and fill out QTI Conversion Forum with "Group#\_CIA#", then for Points Per Question, put in 1. Then format all of your questions and paste them into the box below and click check for syntax error. If there are no syntax errors then click "Convert to canvas QTI file". View "insert the pdf that is labeled (Student\_template\_CIA\_Fall2024\_group.txt)" in order to see an example of a correctly formatted CIA.
  - These are the syntax for each of the following types of questions:
    - **Multiple Choice:** Each choice needs to start with a lower case alphabet, a, b, c, d, etc. with a close parenthesis. The correct choice is designated with an asterisk.
      - 1. What is 2+3?
        - a) 6
        - b) 1
        - \*c) 5
        - d) 10
    - **Multiple-answers:** Each choice use ☐ or ☒ to make the incorrect answers and ☒ for the correct answers.
      - 1. Which of the following are dinosaurs?
        - ☐ Woolly mammoth
        - ☒ Tyrannosaurus rex
        - ☒ Triceratops
        - ☐ Smilodon fatalis

- **Short-answer (fill-in-the-blank):** The correct answers will use an asterisk followed by one or more spaces or tabs followed by an answer.
  1. Who lives at the North Pole?
    - \* Santa
    - \* Santa Claus
    - \* Father Christmas
    - \* Saint Nicholas
    - \* Saint Nick
- **Essay:** This type of question is indicated by a sequence of three or four hashtags.
  1. Write an essay.
 

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- **File-upload:** This type of question is indicated by a sequence of three or four carets.
  1. Upload a file.
 

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- **True/False:** True and False statements start with a) and b). The correct choice is designated with an asterisk.
  1. Water is liquid.
    - \*a) True
    - b) False

## CMAPs - Truman

- What software is used to make CMAPs
  - In order to make CMAPs, you must download the cmap app
  - Downloading cmap
    1. Google "cmaps.ihmc"
    2. Select the first link "Cmap - IHMC" (<https://cmap.ihmc.us/>)
    3. Select orange bar "Downloads"
    4. Select Download CmapTools
    5. Scroll down to the form and fill out questions and boxes- Ignore Donation options at the top of the page (unless you would like to donate) - Tip: choose "Higher Education" for Organization Type. Complete the addition problem (e.g., for "5 + 9" type "14")
    6. Click the correct download button for your computer, "Windows (64 bits)" or "Mac OS X" Tip: If the download does not start automatically, right-click on the download button and select Open in a new window (or tab) or copy the link address and paste into a new browser window. An executable file should download.
    7. Open the executable file (.exe) and follow installation instructions.
- How many have to be done in the year?
  - There must be 4 CMAP's completed before the semester ends. Each CMAP will be due 2 days before each of the Exams at 11:59 pm. This means that CMAP 1

will be due on June, 13 at 11:59 pm, CMAP 2 will be due on June, 20 at 11:59 pm, CMAP 3 will be due on June, 27 at 11:59 pm, and CMAP 4 will be due on July, 5 at 11:59 pm.

- What topics are each done on
  - The cmap may be on any topic covered. This means that CMAP 1 may include arrays, lists, introduction & analysis of algorithms, and doubly linked lists. CMAP 2 may include stacks, bubble sort, selection sort, insertion sort, priority queues, and heaps. CMAP 3 may include hashing, trees, BST, and AVL Trees. CMAP 4 may include divide and conquer sorts, graph, graph traversal, and MST-prims
- How do you upload them, and what files?
  - Once you finish your cmap, save the cmap as a cxi and pdf on your computer as CMAP#(topic). Then, go to the correct cmap assignment in canvas, press submit, and find the cxi and pdf file and submit the cmap.
- How to prompt
  - Prompt gpt with this: "Please provide exhaustive and detailed information on the following aspects of [Topic Name]:
    1. **What**:
      - Provide a definition and key characteristics.
      - Explain the fundamental principles.
      - Discuss the basic structure or components.
      - Include any related terminology and concepts.
    2. **Why**:
      - Mention the advantages and benefits.
      - Explain the problem it solves or the need it addresses.
    3. **Where**:
      - Specify where it is used in the real world.
      - Highlight relevant contexts and scenarios.
    4. **When**:
      - Discuss the situations or conditions for its use.
    5. **How**:
      - Explain the step-by-step processes involved.
      - Include diagrams, examples, and code snippets if applicable.
      - Break down complex processes into manageable steps.
      - Provide detailed explanations for each step or component. "

- After prompting gpt, read and implement each section into cmap
- Example of a cmap
  - This is a link to a completed full credit cmap: (insert Arrays-Phi Tran-Arrays.pdf)

## Dlabs - Kenny

- How do you do them?
  - Dlabs is an abbreviation of Dynamic Labs. They are dynamic in the sense that students have the privilege of choosing which problems they would like to tackle. The link to the problem sets can be found using this link: <https://cses.fi/problemset/>. While Dynamic Labs are an individual assignment for each student, where each student must complete any 3 interview level questions, students are strongly encouraged to collaborate with each other to reach a solution. However, note that students may not submit the exact same solution to any given problem. Solutions must prove that solutions were built collaboratively, not simply given away.
  - **Background on CSES** - The CSES Problem set is a collection of algorithmic programming problems, often regarded as competitive programming problems, designed for testing your understanding of data structures, algorithms, and dynamic programming. The website is a project that originates from Helsinki, Finland.
- What problems can I do? Where do I get the problems from?
  - The following link displays problems from CSES and which topics they relate to, in the order in which the course covers them. Note that the dates reflect a template semester from Fall 2025, but how it's segmented by exam topics is accurate and reliable. <https://markpelico.github.io/cosc2436-schedule/main.html>
  - You are required to create an account on the CSES website in order to test your code and verify that it passes the test cases provided to you. In addition, being signed in allows you to view the success rates of all those who have previously attempted the problem. This can help you gauge the estimated difficulty of the problem.
  - CSES hosts at least 400 problems. Please understand that some problems are inherently simpler, less complex, or in other words, easier, than others. It's also worth mentioning that you may or may not have all the knowledge at your disposal to solve the problem(s) that first catch your interest until later in the semester.
  - If you're reading through an interesting problem that you would like to solve with your own data structures, but are struggling to imagine whether or not a certain data structure could possibly solve it under the time complexity restraints, one option is to consult AI. An example prompt would look like: "I'm currently a student learning data structures and these are the topics I've learned so far \_\_\_\_."

This is the problem I'm trying to solve \_\_\_\_\_. Can I solve this given the time constraints and what I know so far?"

- Keep in mind that data structures can sometimes be interchangeable. This is also where the "Dynamic" portion of Dlabs comes into play. Students have the flexibility to use any tools they have knowledge of to solve the problems, such as using arrays instead of vectors.
- How do I know what I get?
  - You can verify that your solution works if it passes all of the test cases on the CSES website. Note that once you submit, you are able to view the inputs and expected outputs of each test. Each problem will contain tests ensuring that your program passes the time complexity requirements. In addition, you may find that some test cases expect values far larger than any int variable type can handle, which store 32 bits as opposed to the "long long" variable type, which can store 64 bits. Please keep this in mind when your program outputs a value that's drastically off-base from the expected.
- Where do I upload them?
  - To upload your Dlabs, first open FileZilla or any other File Transfer Protocol service you prefer. Then navigate to the remote site and create a folder titled in all lowercase "dlab#\_#". The first # should be the number indicating which set of Dlabs you're submitting (which would be which exam is currently upcoming) and the second # indicates which Dlab problem out of the 3 per exam period you're turning in. Ex: you would create three folders with the following titles: dlab1\_1, dlab1\_2, and dlab1\_3 for submission during the first exam period.
  - Lastly, the only file required to be inside each individual folder is the cpp file containing your coded solution for that problem. So after you drag and drop each file to each of their associated folders, you're done!

## Glabs - Kenny

- How do you do Glabs.
  - There are many ways to find inspiration for these generated lab problems. For instance, you could copy a problem from the CSES website and ask an AI of your choice to modify it in some meaningful way besides just rephrasing the question. Alternatively, you could describe the data structures and constraints and ask AI to create an entirely new problem from scratch. You may use AI tools to help you solve what you've generated, but referencing your solutions to Dlabs covering similar topics is recommended.
- What are Glabs?
  - Glabs are an abbreviation for Generated Labs. Their purpose is to challenge you to create new problems that are suitable for competitions or interviews. Your task is to individually create two unique problems and solve them by any means necessary.
- How do you upload them, and what files do you upload?
  - Once you have two newly generated problem descriptions, including a problem statement with expected results from the program, example input/output, and

time complexity/input size constraints, paste them both into a single Word or Google Doc. Title the document "Glab#", where # refers to which exam period your pair of generated labs covers, and submit that single file to Canvas.

### Grading of the course (just use the one from this semester) - Kenny

- Attendance (1% for lecture + 1% for lab) - 2%
  - Attendance quiz access codes will be distributed in-person and during scheduled class times for lectures and labs.
- Dlab (Individual Submission) - 8%
- CIA (Group Submission) - 3%
- CMAP (Individual Submission) - 3%
- Homework (Individual Submission) - 15%
- Exam 1 - 21%
- Exam 2 - 21%
- Exam 3 (Cumulative Final) - 27%

### How Tests work (ie format, etc...) - Joseph

### Extra Credit opportunities - Kenny

- Exam Challenge
  - The challenge is an opportunity unique to Exam 1. Days before Exam 1, a sign-up form will be released to all students, which they can then decide to opt in or out. To opt in means the following: If you receive a grade of 70 or above on Exam 1, your grade for Exam 1 will be inputted as a 100. However, if you score less than a 70, your grade will be halved (rounding up to the nearest whole number).
  - Please note: the challenge does not modify the contents of Exam 1 in any way. Each student takes the same exam irrespective of whether or not they opted into the challenge. The Exam is NOT made any more difficult for those who choose to take the challenge.
  - For those who fail the challenge, they are offered another opportunity when Exam 2 comes around. If the student earns a 70 or above on Exam 2, then their Exam 1 grade will be raised to a 100, while their Exam 2 grade remains as is. Note that this only applies to Exam 2 if a student opted into the challenge for Exam 1 and failed the challenge.
- Leave rest blank for now. We don't know what extra credit opportunities may show up.



